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June 15, 2026

The Editors  
*Nonlinearity*  
IOP Publishing & London Mathematical Society

Dear Editors,

I am submitting the enclosed manuscript, *Transient Chaos and Topological Bounds in Prime Dynamics: Revisiting the One-Dimensional Sieve Mapping*, for consideration as a research article.

The paper studies the symbolic dynamics of the Eratosthenes sieve under the Metropolis–Stein–Stein (MSS) kneading admissibility framework, taking as starting object the matching, recently introduced in a published companion paper (Wang, *Research in Mathematics* **13**(1), 2026, DOI: 10.1080/27684830.2026.2684334), between the sieve word  $Q_k$  and admissible itineraries of the unimodal map  $x \mapsto 1 - ux^2$  at the band-merging parameter  $u_c \approx 1.5436890$ . Two unconditional theorems anchor the contribution. The *Parity-Gap Lemma* (Theorem A) shows that any MSS-defective shift of  $W_k = Q_k[0, p_{k+1}^2)$  forces a prime gap of length  $\geq p_{k+1} - 1$  at a specific position; this reduces the topological-admissibility question to an extremal prime-gap inequality, and we observe that classical gap bounds (Legendre, Andrica, RH, BHP) all fail to clear it—only a strongly sub-root bound such as Cramér’s conjecture forces eventual admissibility. The *even-gap rigidity* theorem (Theorem B) shows that gaps between consecutive  $L$ -symbols at  $u_c$  are almost surely even, via the parity-twisted lex order on  $RLR^\infty$ . A third structural result (Theorem C) shows the gap measure decays asymptotically geometrically with ratio  $p_\infty \approx 0.596$ , ruling out internal mod-3 resonance and identifying the unimodal projection as an abelian / mod-2 shadow of the prime distribution. The paper closes with a clearly conditional reduction connecting recurrence to the  $L$ - $R$ - $L$  cylinder to a precisely stated arithmetic shadowing problem; we explicitly disclaim any consequence for classical number-theoretic conjectures.

The paper fits squarely within *Nonlinearity*’s remit at the intersection of symbolic dynamics, ergodic theory of one-dimensional unimodal maps, and arithmetic dynamics. The proofs rely on classical results of Misiurewicz, Mañé, Lasota–Yorke, and Young, which are familiar to the journal’s audience. The methodological pattern of the paper—numerical experiments to surface a structural lemma followed by a proof on classical foundations—is in keeping with *Nonlinearity*’s tradition of bridging quantitative experiment and rigorous theory.

I should disclose that I am also the author of the published companion paper cited above, which proposes the unimodal–sieve matching itself; the present paper independently analyses the admissibility structure of that matching and is structured to stand alone—no theorem statement depends on the prior work, as recorded in the title-page footnote. The manuscript is not under consideration elsewhere; there are no competing interests; and the complete LaTeX source, the numerical experiments, the figure-generation scripts, and six companion Jupyter notebooks reproducing the headline numerical claims are openly available at [https://github.com/maris205/prime\\_dynamics](https://github.com/maris205/prime_dynamics).

I thank the Editors and referees in advance for their consideration.

Sincerely yours,

Liang Wang